Status	Finished
Started	Thursday, 13 August 2026, 2:20 PM
Completed	Thursday, 13 August 2026, 2:28 PM
Duration	8 mins 17 secs
Marks	12.00/12.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Complete

Mark 1.00 out of 1.00

Which of the following BEST describes the 'divide-and-conquer' approach in layered network architecture?

- a. Unmanageable tasks are divided into small, manageable tasks to reduce design complexity
- b. Each device handles only its own data without cooperating with others
- c. All network tasks are handled by a single, unified layer to maximize efficiency
- d. Network devices share a single protocol to simplify communication
- e. Data is copied and sent multiple times to improve reliability

The correct answer is:
Unmanageable tasks are divided into small, manageable tasks to reduce design complexity

Question 2

Complete

Mark 1.00 out of 1.00

What key advantage does Modularity provide in a layered network architecture?

- a. It eliminates the need for addressing at each layer.
- b. It forces all layers to use the same protocol.
- c. It requires all layers to be updated simultaneously.
- d. It increases the size of data headers in each layer.
- e. It allows each layer to be modified independently without affecting other layers.

The correct answer is:
It allows each layer to be modified independently without affecting other layers.

Question 3

Complete

Mark 1.00 out of 1.00

The TCP/IP model is described as the 'de facto standard' for network communication. What does 'de facto standard' mean?

a. A standard formally approved and enforced by a government body
b. A standard that became widely accepted through widespread use in practice, not through official legislation
c. A standard that must be licensed before use
d. A temporary standard that is expected to be replaced soon
e. A standard that is only used within academic research environments

The correct answer is:
A standard that became widely accepted through widespread use in practice, not through official legislation

Question 4

Complete

Mark 1.00 out of 1.00

In layered data flow and data encapsulation, what does each layer ADD to the data as it passes DOWNWARD through the sending host?

a. A trailer only
b. A complete copy of the original data
c. Its own header containing information for peer-layer communication
d. Nothing — data passes unchanged through all layers
e. An encryption key for the next layer

The correct answer is:
Its own header containing information for peer-layer communication

Question 5

Complete

Mark 1.00 out of 1.00

What happens to layer headers at the RECEIVING host?

a. They are passed up through all layers and stored permanently
b. Headers are forwarded to the next hop without any modification
c. They are deleted by the physical layer before reading
d. Each layer removes the header added by its corresponding peer layer at the sender.
e. Only the application layer removes all headers at once

The correct answer is:
Each layer removes the header added by its corresponding peer layer at the sender.

Question 6

Complete

Mark 1.00 out of 1.00

At the Network Layer, when a packet travels from sender to recipient through multiple routers, which addresses CHANGE and which remain the SAME?

- a. MAC addresses change at each hop, but source and destination IP addresses remain the same
- b. Only the destination IP address changes; the source IP stays the same
- c. Both MAC addresses and IP addresses remain constant throughout the journey
- d. IP addresses change at each router, but MAC addresses remain the same
- e. Both MAC and IP addresses change at every router hop

The correct answer is:
MAC addresses change at each hop, but source and destination IP addresses remain the same

Question 7

Complete

Mark 1.00 out of 1.00

Which of the following correctly describes the SCOPE of communication for Layers 2, 3, and 4?

- a. L2: process-to-process, L3: end-to-end, L4: hop-to-hop
- b. L2: end-to-end, L3: hop-to-hop, L4: host-to-host
- c. L2: hop-to-hop, L3: host-to-host (source to destination), L4: end-to-end (process-to-process)
- d. L2: source to destination, L3: end-to-end, L4: hop-to-hop
- e. L2: host-to-host, L3: hop-to-hop, L4: process-to-process

The correct answer is:
L2: hop-to-hop, L3: host-to-host (source to destination), L4: end-to-end (process-to-process)

Question 8

Complete

Mark 1.00 out of 1.00

What is the primary purpose of a layer header?

- a. Replace all addressing information contained in lower-layer headers
- b. Carry information required to perform that layer's functions
- c. Control the internal electronic circuits of every network device
- d. Increase the physical transmission speed of the communication medium
- e. Store application files before they enter the communication system

The correct answer is: Carry information required to perform that layer's functions

Question 9

Complete

Mark 1.00 out of 1.00

Which statement correctly compares Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM)?

a. TDM separates users by frequency bands, while FDM separates users by time slots.

b. TDM separates users by time slots, while FDM separates users by frequency bands.

c. TDM and FDM both assign different IP addresses to distinguish communication channels.

d. TDM is used only for analog data, while FDM is used only for digital data.

e. TDM requires separate cables, while FDM requires separate network addresses.

The correct answer is: TDM separates users by time slots, while FDM separates users by frequency bands.

Question 10

Complete

Mark 1.00 out of 1.00

A 48-bit MAC address can normally be represented using:

a. Sixteen hexadecimal digits

b. Eight hexadecimal digits

c. Ten hexadecimal digits

d. Thirty-two hexadecimal digits

e. Twelve hexadecimal digits

The correct answer is: Twelve hexadecimal digits

Question 11

Complete

Mark 1.00 out of 1.00

Which command is presented in the lecture for displaying detailed Windows network configuration?

a. tracert

b. netstat -rn

c. ifconfig -a

d. ping

e. ipconfig /all

The correct answer is: ipconfig /all

Question 12

Complete

Mark 1.00 out of 1.00

In the layered addressing model, which set CORRECTLY maps each address type to its correct size?

a. MAC = 48-bit, IPv4 = 32-bit, Port = 16-bit

b. MAC = 32-bit, IP = 48-bit, Port = 16-bit

c. MAC = 48-bit, IPv4 = 16-bit, Port = 32-bit

d. MAC = 64-bit, IPv4 = 32-bit, Port = 8-bit

e. MAC = 32-bit, IPv4 = 64-bit, Port = 8-bit

The correct answer is: MAC = 48-bit, IPv4 = 32-bit, Port = 16-bit